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S11-07

Aim: Write a program to implement Stack and Queue using arrays.

Code (Stack):

```
#include<stdio.h>

int stack[100], i, choice=0, top=-1,n;

void main()
{
    printf("Enter the number of elements in the stack \n");
    scanf("%d",&n);
    while(choice!=4)
    {
        printf("\n Choose one of the following choices:\n");
        printf("\n 1.PUSH");
        printf("\n 2.POP");
        printf("\n 3.SHOW");
        printf("\n 4.EXIT");

        printf("\n Enter the choice:\n");
        scanf("%d",&choice);
        switch(choice)
        {
            case 1: push();
                break;

            case 2: pop();
                break;

            case 3: show();
                break;
```

```

        case 4: printf("Exit");
        break;

        default: printf("invalid choice!!");
    }
}
}

void push(int val, int n)
{
    if(top==n)
    {
        printf("\n Overflow");
    }
    else
    {
        printf("Enter the value:");
        scanf("%d", &val);
        top=top+1;
        stack[top]=val;
    }
}

int pop()
{
    if(top== -1)
    {
        printf("\n Underflow");
        return 0;
    }
}

```

```

else
{

    top=top-1;
    return stack[top];
}
}

void show()
{
    for(i=top;i>=0; i--)
    {
        printf("%d\t", stack[i]);
    }
    if(top== -1)
    {
        printf("Stack is empty");
    }
}

```

Output (Stack) :

Enter the number of elements in the stack

4

Choose one of the following choices:

- 1.PUSH
- 2.POP
- 3.SHOW
- 4.EXIT

Enter the choice:

1

Enter the value:1

Choose one of the following choices:

- 1.PUSH
- 2.POP
- 3.SHOW
- 4.EXIT

Enter the choice:

1

Enter the value:2

Choose one of the following choices:

- 1.PUSH
- 2.POP
- 3.SHOW
- 4.EXIT

Enter the choice:

3

2 1

Choose one of the following choices:

- 1.PUSH
- 2.POP
- 3.SHOW
- 4.EXIT

Enter the choice:

2

Choose one of the following choices:

- 1.PUSH
- 2.POP
- 3.SHOW
- 4.EXIT

Enter the choice:

3

1

Choose one of the following choices:

1.PUSH

2.POP

3.SHOW

4.EXIT

Enter the choice:

4

Exit

Code (Queue):

```
#include <stdio.h>
```

```
int queue[100];
```

```
void main()
```

```
{
```

```
    int choice=0,size;
```

```
    printf("Enter the number of elements in queue: ");
```

```
    scanf("%d",&size);
```

```
    int front=-1,rear=-1;
```

```
    while(choice<4)
```

```
    {
```

```
        printf("\n1.Enqueue\n2.Dequeue\n3.Display\n4.Exit\nEnter your choice: ");
```

```
        scanf("%d",&choice);
```

```
        switch(choice)
```

```
        {
```

```
            case 1:
```

```
                enqueue(&front,&rear,size);
```

```
                break;
```

case 2:

 dequeue(&front,&rear,size);

 break;

case 3:

 display(&front,&rear);

 break;

case 4:

 printf("Exiting..");

 break;

default:

 printf("invalid choice");

 break;

}

}

}

void enqueue(int*p, int*q, int len)

{

 int value;

 if(*q==len-1)

 {

 printf("\nOverflow\n");

 }

 else

 {

 if((*p)==-1)

 {

 *p=0;

 }

```

        printf("\nEnter the value you want to insert to the queue: ");
        scanf("%d",&value);
        (*q)++;
        queue[*q]=value;
    }
}
void dequeue(int *p,int*q,int len)
{
    if(*p== -1 || *p>*q)
    {
        printf("\nUnderflow\n");
    }
    else
    {
        (*p)++;
    }
}
void display(int*p,int*q)
{
    int i;
    if(*p== -1 || *p>*q)
    {
        printf("Queue is empty\n");
    }
    else
    {
        for(i=*p;i<=*q;i++)
        {
            printf("%d\n",queue[i]);

```

```
}  
}  
}
```

Output (Queue) :

Enter the number of elements in queue: 4

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter your choice: 1

Enter the value you want to insert to the queue: 1

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter your choice: 1

Enter the value you want to insert to the queue: 2

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter your choice: 3

1

2

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter your choice: 2

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter you choice: 3

2

1.Enqueue

2.Dequeue

3.Display

4.Exit

Enter your choice: 4

Exiting..